A Extracellular electrophysiology (ecephys)

```
microephys_toy/
├── dataset_description.json
├── participants.tsv
├── probes/
│   └── customprobe1.json
├── sub-mouse01/
│   ├── ecephys/
│   │   ├── sub-mouse01_probes.tsv
│   │   ├── sub-mouse01_probes.json
│   │   ├── sub-mouse01_electrodes.tsv
│   │   ├── sub-mouse01_channels.tsv
│   │   ├── sub-mouse01_space-AllenCCFv3_electrodes.tsv
│   │   └── sub-mouse01_space-AllenCCFv3_coordsystem.json
│   └── ses-01/ecephys/
│       ├── …_task-reach_ecephys.json
│       ├── …_task-reach_ecephys.nwb/nix
│       ├── …_task-reach_events.tsv
│       ├── …_task-rest_ecephys.json
│       └── …_task-rest_ecephys.nwb
```

***_probes.tsv** *Physical recording devices*

probe_name	type	manufacturer	model	electrode_count	hemisphere	associated_brain_region
probe01	silicon-probe	NeuroNexus	A1x32-Poly3-10mm-50-177	32	L	Primary Motor Cortex
probe02	silicon-probe	Custom Lab	customprobe1	16	R	Hippocampus CA1

***_electrodes.tsv** *Probe-relative contact positions*

name	probe_name	x	y	z	impedance	shank_id	material	location
e001	probe01	0	0	0	1.2	0	iridium-oxide	MOp
e002	probe01	0	0	25	1.1	0	iridium-oxide	MOp
⋮								
e033	probe02	0	0	0	1.5	0	iridium-oxide	CA1

***_channels.tsv** *Recording channel properties*

name	electrode_name	type	units	sampling_frequency	gain	low_cutoff	high_cutoff	status
ch001	e032	LFP	uV	1000	500	0.1	300	good
ch017	e032	HP	uV	30000	500	300	10000	good
ch033	e048	LFP	uV	1000	500	0.1	300	good
sync01	n/a	SYNC	V	30000	1	n/a	n/a	good

***_ecephys.json**

```
{ "TaskName": "Reach to Grasp",
  "SamplingFrequency": 30000,
  "Manufacturer": "IMEC",
  "SampleEnvironment": "in vivo", … }
```

***_coordsystem.json**

```
{ "MicroephysCoordinateSystem": "AllenCCFv3",
  "MicroephysCoordinateUnits": "um",
  "MicroephysCoordinateSystemDescription":
    "Origin at anterior commissure…" }
```

B Intracellular electrophysiology (icephys)

```
└── sub-mouse02/
    ├── sub-mouse02_scans.tsv
    └── icephys/
        ├── sub-mouse02_probes.tsv
        ├── sub-mouse02_electrodes.tsv
        ├── sub-mouse02_channels.tsv
        ├── …_task-IVcurve_icephys.json
        ├── …_task-IVcurve_icephys.nwb/nix
        └── …_task-IVcurve_events.tsv
```

***_probes.tsv** *Pipette devices*

probe_name	type	manufacturer	electrode_count	hemisphere	associated_brain_region
pipette01	patch-pipette	Sutter	1	L	Visual Cortex Layer 2/3
pipette02	patch-pipette	Sutter	1	R	Visual Cortex Layer 2/3

***_electrodes.tsv** *Pipette-specific properties*

name	probe_name	pipette_solution	internal_pipette_diameter	external_pipette_diameter	material
patch01	pipette01	K-gluconate	1.5	2.5	borosilicate-glass
patch02	pipette02	K-gluconate	1.5	2.5	borosilicate-glass

***_channels.tsv** *Recording channels with mode*

name	type	units	sampling_frequency	recording_mode	gain	ground	status
patch01_vm	VM	mV	20000	current-clamp	10	AgCl	good
patch01_im	IM	pA	20000	current-clamp	5	AgCl	good
patch02_vm	VM	mV	20000	voltage-clamp	10	AgCl	good
patch02_im	IM	pA	20000	voltage-clamp	5	AgCl	good

***_icephys.json**

```
{ "TaskName": "IV Curve
Characterization",
  "SamplingFrequency": 20000,
  "Manufacturer": "Molecular Devices",
  "SampleEnvironment": "ex vivo", … }
```